

CISP 400

OOP in C++

Dr. Fowler



F O L S O M L A K E C O L L E G E
EL DORADO CENTER | RANCHO CORDOVA CENTER

Agenda

Ch 10 - Pointers

C++ Memory Model

10.1 Pointers and the Address Operator

10.2 Pointer Variables

10.3 The Relationship between Arrays and pointers

10.4 Pointer Arithmetic

10.5 Initializing Pointers

10.6 Comparing Pointers

10.7 Pointers as Function Parameters

10.8 Pointers to Constants and Constant Pointers

10.9 Dynamic Memory Allocation

10.10 Returning Pointers from Functions

~~10.11 Pointers to Class Objects and Structures~~

~~10.12 Selecting Members of Objects~~

10.13 Smart Pointers

Agenda

Ch 10 - Pointers

Code Dynamic Memory

Personal Review of
Homework 1

Programming Exercise

- A local retailer has asked you to help put together a program to track customers.
- Create a menu with the following options: add customer, remove customer, display all customers, quit.
- Create a dynamic array holding customer name. Add to the array with the add customer option, delete from the array with the remove customer option (just the last customer added for now).

Programming Exercise

- Modularize (put code in functions) this program. Pay special attention to the dynamic memory. If you made the dynamic array global, make it local now and put it in main().
- Using any technique you know of add int customer number to your data structure.
- Rewrite the remove customer option to prompt for the customer number and then remove that customer (rather than the last one).

Programming Exercise

- If you have time:
 - add input validation for the menu choice.
 - Do not add a blank name.
 - Detect a delete with 0 customers.
 - Generate a warning if there are no customers to display.
 - Add a capacity function. Instead of changing the array for each add and delete, double the size of the array when it is full and reduce the size of the array when the number of elements are less than 50%. Capacity is the number of available elements, size is the number of populated elements.